

Arancha Ramírez

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EDUCATION

- **Imperial College London - Faculty of Engineering** London, UK
MEng Design Engineering 2021-2026
- **British Council** Madrid, Spain
Spanish Baccalaureate, IGCSE's 2012-2021

PROFILE

Engineering Graduate with hands-on experience in data-driven decision support (RO, Monte Carlo Simulation, MCDA), forecasting and machine-learning modelling and workflow automation (n8n). Through academic projects I have built algorithms in Python, MATLAB and R for data processing, analysis and automated testing, and translated technical results into clear, business-facing reports and tools for both specialists and non-specialists stakeholders. I am highly motivated to apply this analytical, business-oriented foundation to the technology and consulting practice.

WORK EXPERIENCE

- **Strategic Engineering Consulting Ltd** London, UK
Strategic Engineering Analyst Intern April 2025-June 2026
 - Collaborated on the firm's partnership with ZUKEN, developing frameworks to embed strategic engineering principles into Model-Based Systems Engineering practice.
 - Built tailored AI agents and automated workflows using n8n, streamlining internal operations and automating content generation pipelines.
 - Led SEO optimisation, web design, and UI/UX improvements, enhancing digital visibility and user engagement.
- **GeneracionCode** Madrid, Spain
Program Development Coordinator August 2024
 - Developed and pitched partnership proposals to major companies including ITP Aero, securing sponsorship for children's programming courses.
 - Designed programming education initiatives fostering coding skills and digital literacy in young learners, aligned with corporate social responsibility goals.

ACADEMIC PROJECTS

- **Strategic Engineering Analysis of Desalination Investments under Uncertainty** London, UK
Final Year Master's Project October 2025 - June 2026
 - Built an integrated and scalable decision-support framework combining real-options, robustness, and multi-criteria decision analysis to value managerial flexibility in modular desalination investments under uncertainty.
 - Calibrated stochastic models of water and electricity price uncertainty; ran paired Monte Carlo trials over investment horizon, quantifying the Expected Value of Flexibility, with the flexible design outperforming the rigid design and remaining positive under perturbations.
 - Designed and developed an interactive web-based Decision-Support Tool exposing distributions, weight-sensitivity, and trade-offs to both specialists and non-specialists stakeholders.
- **WasteLens - Plastic waste as a global trade currency** London, UK
Data to Product Module October - December 2025
 - Sourced and curated waste trade records, designing an imputation method, using country-specific smoothed unit prices to fill in gaps in reported trade quantities.
 - Evaluated ARIMA against Exponential Smoothing for forecasting volumes, applied bottom up ARIMA model and designed the interactive interface translating the underlying models.
- **Sustainable intervention to reduce Foodji's carbon footprint** London, UK
Sustainable Design & Strategy Module January - March 2026
 - Identified future drivers of sustainable transition and designed a transparency and incentivisation intervention with a supporting business case, quantifying CO₂ reduction, economic viability and user retention to underpin feasibility analysis.

SKILLS

- **Engineering & Analytical:** systems modelling, financial modelling, Monte Carlo Simulation, Real-options analysis, MCDA, time-series forecasting, Ahrefs, Google Search Console, n8n, Ansys FEA
- **Programming languages:** Python, JavaScript, MATLAB, C++, R (incl. Shiny)
- **CAD & Design:** SolidWorks, Fusion 360, AutoCAD, Blender, Keyshot, Indesign, Adobe, Figma, Framer
- **Office:** Word, PowerPoint, Excel
- **Languages:** Proficient in English, Spanish and French, Chinese (IGCSE)